

Trigonometry

1. Trigonometry

1.1. Domain/Range of trig functions

Identities	Domain	Range
$\sin \theta$	$\mathbb{R}$	$[-1, 1]$
$\cos \theta$	$\mathbb{R}$	$[-1, 1]$
$\tan \theta$	$\mathbb{R} - \{(2n + 1)\frac{\pi}{2}\}$	$\mathbb{R}$
$\cot \theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R}$
$\csc \theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R} - (-1, 1)$
$\sec \theta$	$\mathbb{R} - \{(2n + 1)\frac{\pi}{2}\}$	$\mathbb{R} - (-1, 1)$

1.2. Trigonometric Formulas

$$1. \sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$2. \cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$3. \tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

$$4. \cot \theta = \frac{\text{Adjacent}}{\text{Opposite}}$$

$$5. \sec \theta = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

$$6. \csc \theta = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

$$7. \sin \theta = \frac{1}{\csc \theta}$$

$$8. \cos \theta = \frac{1}{\sec \theta}$$

$$9. \tan \theta = \frac{1}{\cot \theta} = \frac{\sin \theta}{\cos \theta}$$

$$10. \cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$$

$$11. \sec \theta = \frac{1}{\cos \theta}$$

$$12. \csc \theta = \frac{1}{\sin \theta}$$

$$13. \sin(-x) = -\sin x$$

$$14. \cos(-x) = \cos x$$

$$15. \tan(-x) = -\tan x$$

$$16. \cot(-x) = -\cot x$$

$$17. \sec(-x) = \sec x$$

$$18. \csc(-x) = -\csc x$$

$$19. \sin^2 x + \cos^2 x = 1$$

$$20. \sec^2 x - \tan^2 x = 1$$

$$21. \csc^2 x - \cot^2 x = 1$$

$$22. \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$23. \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$24. \tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$25. \cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$$

$$26. \quad 2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$27. \quad 2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$28. \quad 2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$29. \quad 2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$30. \quad \sin C + \sin D = 2 \sin\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$$

$$31. \quad \sin C - \sin D = 2 \cos\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{C - D}{2}\right)$$

$$32. \quad \cos C + \cos D = 2 \cos\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$$

$$33. \quad \cos C - \cos D = 2 \sin\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{D - C}{2}\right)$$

$$34. \quad \sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$35. \quad \cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$36. \quad \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$37. \quad \cot 2A = \frac{\cot^2 A - 1}{2 \cot A}$$

$$38. \quad \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$39. \quad \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$40. \quad \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$41. \quad \cot 3A = \frac{\cot^3 A - 3 \cot A}{3 \cot^2 A - 1}$$

$$42. \quad \begin{aligned} \sin(A + B) \cdot \sin(A - B) &= \sin^2 A - \sin^2 B \\ &= \cos^2 B - \cos^2 A \end{aligned}$$

$$43. \quad \begin{aligned} \cos(A + B) \cdot \cos(A - B) &= \cos^2 A - \sin^2 B \\ &= \cos^2 B - \sin^2 A \end{aligned}$$

$$44. \tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$$

1.3. Value of θ

$$1. \sin \theta = \sin \alpha \implies \theta = n\pi + (-1)^n \alpha$$

$$2. \cos \theta = \cos \alpha \implies \theta = 2n\pi \pm \alpha$$

$$3. \tan \theta = \tan \alpha \implies \theta = n\pi + \alpha$$

$$4. \sin \theta = 0 \implies \theta = n\pi$$

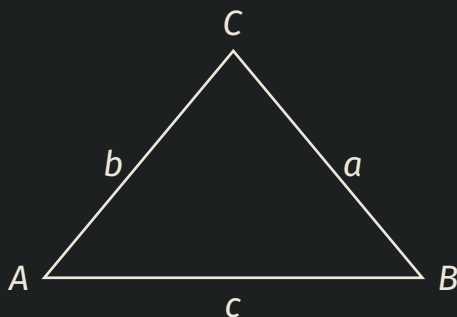
$$5. \cos \theta = 0 \implies \theta = n\pi + \frac{\pi}{2} = (2n + 1)\frac{\pi}{2}$$

$$6. \tan \theta = 0 \implies \theta = n\pi$$

$$7. \sin \theta = 1 \implies \theta = 2n\pi + \frac{\pi}{2} = (4n + 1)\frac{\pi}{2}$$

$$8. \cos \theta = 1 \implies \theta = 2n\pi$$

2. Properties of a triangles



2.1. Area of a triangle

$$1. \Delta = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B$$

$$2. \Delta = \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}$$

2.2. Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where circumradius, $R = \frac{abc}{4\Delta}$

2.3. Law of Cosines:

$$1. \ a^2 = b^2 + c^2 - 2bc \cos A \implies \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$2. \ b^2 = a^2 + c^2 - 2ac \cos B \implies \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$3. \ c^2 = a^2 + b^2 - 2ab \cos C \implies \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

2.4. Projection Rule:

$$1. \ a = b \cos C + c \cos B$$

$$2. \ b = c \cos A + a \cos C$$

$$3. \ c = a \cos B + b \cos A$$

2.5. Napier's analogy:

$$1. \ \tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$2. \ \tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2}$$

$$3. \ \tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}$$

2.6. Half Angle Formulas:

$$1. \ \sin\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$2. \cos\left(\frac{A}{2}\right) = \sqrt{\frac{s(s-a)}{bc}}$$

$$3. \tan\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{r}{(s-a)}$$

$$\text{where inradius, } r = 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right)$$

3. Inverse Trigonometry

3.1. Domain/Range of inverse trig functions

Identities	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$

3.2. Inverse Trigonometric Formulas

$$1. \sin^{-1} x = \csc^{-1}\left(\frac{1}{x}\right)$$

$$2. \cos^{-1} x = \sec^{-1}\left(\frac{1}{x}\right)$$

$$3. \tan^{-1} x = \cot^{-1}\left(\frac{1}{x}\right), x > 0$$

$$= \cot^{-1}\left(\frac{1}{x}\right) - \pi, x < 0$$

$$= 0, x = 0$$

$$4. \csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \sec^{-1} x$$

5. $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \csc^{-1} x$
6. $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1}(x)$
7. $\sin^{-1}(-x) = -\sin^{-1} x$
8. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
9. $\tan^{-1}(-x) = -\tan^{-1} x$
10. $\cot^{-1}(-x) = \pi - \cot^{-1} x$
11. $\csc^{-1}(-x) = -\csc^{-1} x$
12. $\sec^{-1}(-x) = \pi - \sec^{-1} x$
13. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
14. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
15. $\csc^{-1} x + \sec^{-1} x = \frac{\pi}{2}$
16. $\sin^{-1} x \pm \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} \pm y\sqrt{1-x^2}\right)$
17. $\cos^{-1} x \pm \cos^{-1} y = \cos^{-1}\left(xy \mp \sqrt{1-x^2}\sqrt{1-y^2}\right)$
18. $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1$
 $= \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy > 1, x > 0, y > 0$
 $= -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy > 1, x < 0, y < 0$
19. $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$
20. $\tan^{-1} x \pm \tan^{-1} y \pm \tan^{-1} z = \tan^{-1}\left(\frac{x \pm y \pm z \mp xyz}{1 \mp xy \mp yz \mp zx}\right)$
21. $2 \sin^{-1} x = \cos^{-1}(1-2x^2) = \sin^{-1}(2x\sqrt{1-x^2})$
22. $2 \cos^{-1} x = \cos^{-1}(2x^2-1)$
23. $2 \tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$
24. $3 \sin^{-1} x = \sin^{-1}(3x-4x^3)$

$$25. \quad 3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x)$$

4. Hyperbolic Trigonometry

4.1. Domain/Range of Hyperbolic Functions

Identities	Domain	Range
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$
$\cosh x$	$\mathbb{R}$	$[1, \infty)$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$
$\coth x$	$\mathbb{R} - \{0\}$	$(-\infty, -1) \cup (1, \infty)$
$\operatorname{csch} x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$

4.2. Hyperbolic Trigonometric Formulas

$$1. \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

$$2. \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$3. \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$4. \quad \coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$5. \quad \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$6. \quad \operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

$$7. \quad e^x = \cosh x + \sinh x$$

$$8. \quad e^{-x} = \cosh x - \sinh x$$

$$9. \quad \sinh \theta = \frac{1}{\operatorname{csch} \theta}$$

$$10. \quad \cosh \theta = \frac{1}{\operatorname{sech} \theta}$$

$$11. \tanh \theta = \frac{1}{\coth \theta} = \frac{\sinh \theta}{\cosh \theta}$$

$$12. \coth \theta = \frac{1}{\tanh \theta} = \frac{\cosh \theta}{\sinh \theta}$$

$$13. \operatorname{sech} \theta = \frac{1}{\cosh \theta}$$

$$14. \operatorname{csch} \theta = \frac{1}{\sinh \theta}$$

$$15. \sinh(-x) = -\sinh x$$

$$16. \cosh(-x) = \cosh x$$

$$17. \tanh(-x) = -\tanh x$$

$$18. \coth(-x) = -\coth x$$

$$19. \operatorname{sech}(-x) = \operatorname{sech} x$$

$$20. \operatorname{csch}(-x) = -\operatorname{csch} x$$

$$21. \cosh^2 x - \sinh^2 x = 1$$

$$22. \operatorname{sech}^2 x + \tanh^2 x = 1$$

$$23. \coth^2 x - \operatorname{csch}^2 x = 1$$

$$24. \sinh(A \pm B) = \sinh A \cosh B \pm \cosh A \sinh B$$

$$25. \cosh(A \pm B) = \cosh A \cosh B \pm \sinh A \sinh B$$

$$26. \tanh(A \pm B) = \frac{\tanh A \pm \tanh B}{1 \pm \tanh A \tanh B}$$

$$27. \coth(A \pm B) = \frac{\coth A \coth B \pm 1}{\coth B \pm \coth A}$$

$$28. \sinh 2A = 2 \sinh A \cosh A = \frac{2 \tanh A}{1 - \tanh^2 A}$$

$$29. \cosh 2A = \cosh^2 A + \sinh^2 A = 2 \cosh^2 A - 1 = 1 + 2 \sinh^2 A = \frac{1 + \tanh^2 A}{1 - \tanh^2 A}$$

$$30. \tanh 2A = \frac{2 \tanh A}{1 + \tanh^2 A}$$

$$31. \coth 2A = \frac{\coth^2 A + 1}{2 \coth A}$$

$$32. \sinh\left(\frac{A}{2}\right) = \pm \sqrt{\frac{\cosh A - 1}{2}}$$

$$33. \cosh\left(\frac{A}{2}\right) = \sqrt{\frac{\cosh A + 1}{2}}$$

$$34. \tanh\left(\frac{A}{2}\right) = \frac{\cosh A - 1}{\sinh A} = \frac{\sinh A}{\cosh A + 1} = \sqrt{\frac{\cosh A - 1}{\cosh A + 1}}$$

$$35. \sinh 3A = 3 \sinh A + 4 \sinh^3 A$$

$$36. \cosh 3A = 4 \cosh^3 A - 3 \cosh A$$

$$37. \tanh 3A = \frac{3 \tanh A + \tanh^3 A}{1 + 3 \tanh^2 A}$$

$$38. 2 \sinh A \cosh B = \sinh(A + B) + \sinh(A - B)$$

$$39. 2 \cosh A \sinh B = \sinh(A + B) - \sinh(A - B)$$

$$40. 2 \cosh A \cosh B = \cosh(A + B) + \cosh(A - B)$$

$$41. 2 \sinh A \sinh B = \cosh(A + B) - \cosh(A - B)$$

$$42. \sinh C + \sinh D = 2 \sinh\left(\frac{C + D}{2}\right) \cdot \cosh\left(\frac{C - D}{2}\right)$$

$$43. \sinh C - \sinh D = 2 \cosh\left(\frac{C + D}{2}\right) \cdot \sinh\left(\frac{C - D}{2}\right)$$

$$44. \cosh C + \cosh D = 2 \cosh\left(\frac{C + D}{2}\right) \cdot \cosh\left(\frac{C - D}{2}\right)$$

$$45. \cosh C - \cosh D = 2 \sinh\left(\frac{C + D}{2}\right) \cdot \sinh\left(\frac{C - D}{2}\right)$$

$$46. (\cosh x \pm \sinh x)^n = \cosh(nx) \pm \sinh(nx) = e^{\pm nx}$$

5. Inverse Hyperbolic Functions

5.1. Domain/Range & Logarithmic Forms

Identities	Domain	Range
$\sinh^{-1} x$	$\mathbb{R}$	$\mathbb{R}$
$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$
$\tanh^{-1} x$	$(-1, 1)$	$\mathbb{R}$
$\coth^{-1} x$	$(-\infty, -1) \cup (1, \infty)$	$\mathbb{R} - \{0\}$
$\operatorname{sech}^{-1} x$	$(0, 1]$	$[0, \infty)$
$\operatorname{csch}^{-1} x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$

5.2. Inverse Hyperbolic Trigonometric Formulas

1. $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$
2. $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1$
3. $\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$
4. $\coth^{-1} x = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right), \quad |x| > 1$
5. $\operatorname{sech}^{-1} x = \ln\left(\frac{1 + \sqrt{1 - x^2}}{x}\right), \quad 0 < x \leq 1$
6. $\operatorname{csch}^{-1} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right), \quad x \neq 0$
7. $\sinh^{-1}\left(\frac{1}{x}\right) = \operatorname{csch}^{-1} x, \quad x \neq 0$
8. $\cosh^{-1}\left(\frac{1}{x}\right) = \operatorname{sech}^{-1} x, \quad 0 < x \leq 1$
9. $\tanh^{-1}\left(\frac{1}{x}\right) = \coth^{-1} x, \quad |x| > 1$
10. $\coth^{-1}\left(\frac{1}{x}\right) = \tanh^{-1} x, \quad 0 < |x| < 1$
11. $\operatorname{sech}^{-1}\left(\frac{1}{x}\right) = \cosh^{-1} x, \quad x \geq 1$
12. $\operatorname{csch}^{-1}\left(\frac{1}{x}\right) = \sinh^{-1} x, \quad x \neq 0$
13. $\sinh^{-1}(-x) = -\sinh^{-1} x$
14. $\cosh^{-1}(-x) = \pi - \cosh^{-1} x$
15. $\tanh^{-1}(-x) = -\tanh^{-1} x$
16. $\coth^{-1}(-x) = -\coth^{-1} x$
17. $\operatorname{sech}^{-1}(-x) = \pi - \operatorname{sech}^{-1} x$
18. $\operatorname{csch}^{-1}(-x) = -\operatorname{csch}^{-1} x$
19. $\sinh^{-1} x \pm \sinh^{-1} y = \sinh^{-1}\left(x\sqrt{1+y^2} \pm y\sqrt{1+x^2}\right)$

$$20. \cosh^{-1} x \pm \cosh^{-1} y = \cosh^{-1} \left(xy \pm \sqrt{(x^2 - 1)(y^2 - 1)} \right)$$

$$21. \tanh^{-1} x + \tanh^{-1} y = \tanh^{-1} \left(\frac{x + y}{1 + xy} \right)$$

$$22. \tanh^{-1} x - \tanh^{-1} y = \tanh^{-1} \left(\frac{x - y}{1 - xy} \right)$$

$$23. 2 \sinh^{-1} x = \cosh^{-1} (2x^2 + 1)$$

$$24. 2 \cosh^{-1} x = \cosh^{-1} (2x^2 - 1)$$

$$25. 2 \tanh^{-1} x = \tanh^{-1} \left(\frac{2x}{1 - x^2} \right) = \sinh^{-1} \left(\frac{2x}{1 - x^2} \right) = \cosh^{-1} \left(\frac{1 + x^2}{1 - x^2} \right)$$